# Práctica tres: Sistema musculoesquelético

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## Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

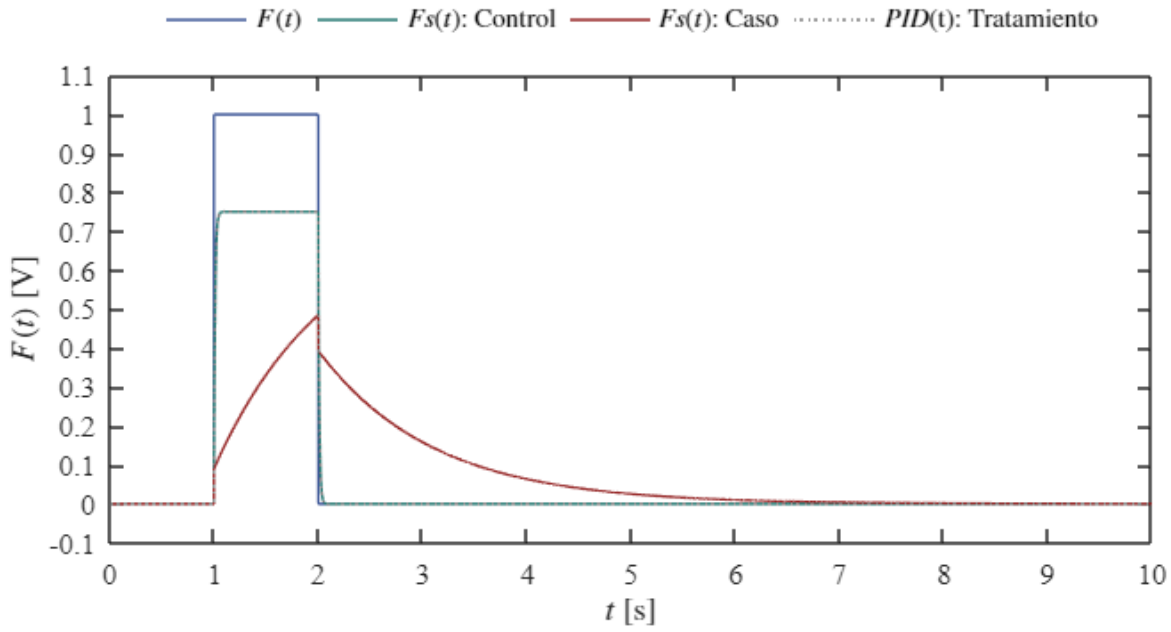
Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

## Datos de la simulación

```
clc; clear; close all; warning('off','all')
tend = '10';
open_system('sistema4')
parameters.StopTime = '10';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
parameters.TimeSaveName = 't';
x = sim('sistema4',parameters);
```

## Respuesta: Control y caso

```
plotsignals(x.t,x.F,x.Fsx,x.Fsy,x.PID,'Control y caso')
```



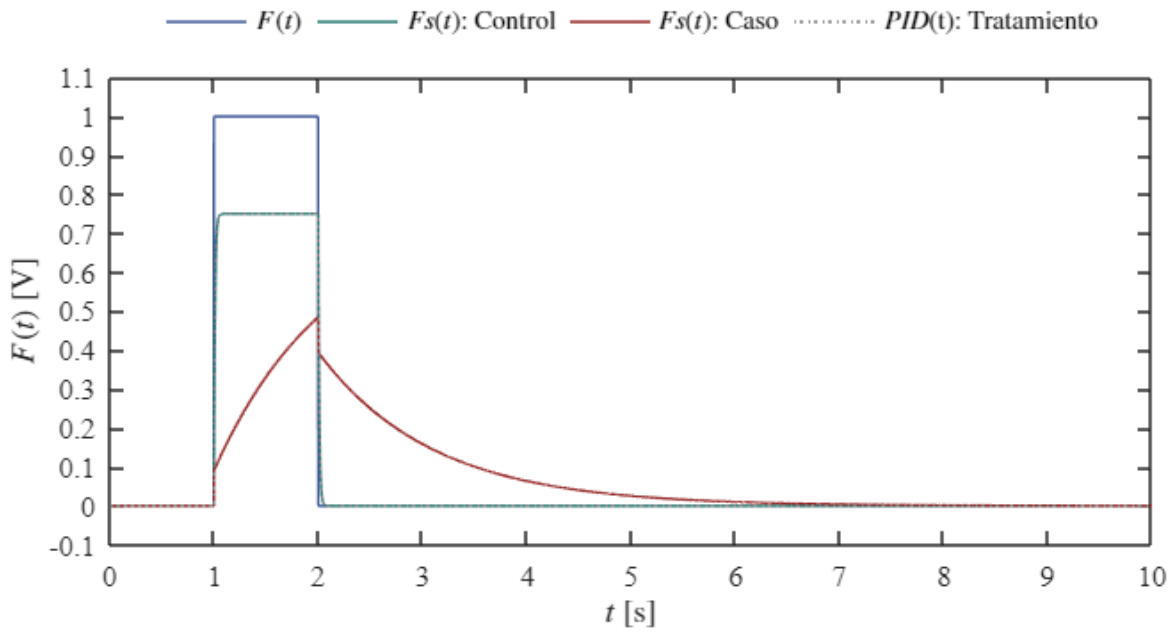

## Modelo de EDOs

```
dt = 1e-3;
t = (0:dt:10);
F = zeros(length(t),1); F(1/dt:2/dt) = 1;
PID = interp1(x.t, x.PID, t, 'previous');

par_control = [0.25,10e-6,100e-6,100];
par_caso = [0.25,10e-6,100e-6,10e3];

Fs_control = musculoesqueletico(par_control,F,t);
Fs_caso = musculoesqueletico(par_caso,F,t);
```

```
plotsignals(t,F,Fs_control,Fs_caso,PID,'EDOsys')
```




```
function Fs = musculoesqueletico(par,F,t)
    alpha = par(1); Cs = par(2); Cp = par(3); R = par(4); dt = t(2);
    n = round(t(end)/dt);
    x = zeros(length(t),1); x(1) = 0;
    Fs = zeros(length(t),1); Fs(1) = 0;
    for i = 1:n
        x(i+1) = x(i) + ((F(i)-x(i))/(R*(Cp+Cs)))*dt; %EDO
        Fs(i+1) = Cs*F(i)/(Cp+Cs) + (1 - alpha - Cs/(Cp+Cs))*x(i); %Función
    end
end
```

## Función: Respuesta de la simulación

```
function plotsignals(t,F,Fs1,Fs2,pid,name)
    set (figure(),'color','w')
    set (gcf,'units','inches','position',[1,1,7,3.5])
    set (gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;

    clr1 = [66, 91, 154]/255;
    clr2 = [40, 122, 116]/255;
    clr3 = [151, 40, 40]/255;
    clr4 = [0.5, 0.5, 0.5];

    plot (t,F,'-','LineWidth',0.25,'Color',clr1)
    plot (t,Fs1,'-','LineWidth',0.25,'Color',clr2)
```

```
    plot (t,Fs2,'-','LineWidth',0.25,'Color',clr3)
    plot (t,pid,':','LineWidth',0.25,'Color',clr4)

    L = legend('$F(t)$','$Fs(t)$: Control','$Fs(t)$: Caso','$PID$(t): Tratamiento');

set(L,'interpreter','latex','location','northoutside','orientation','horizontal','bo
x','off')
    xlabel('$t$ [s]','interpreter','latex')
    ylabel('$F(t)$ [V]','interpreter','latex')

    xlim([0 10]); xticks(0:1:15)
    ylim([-0.1 1.1]); yticks(-0.1:0.1:1.1)

    exportgraphics(gcf,['matlab_',name,'.pdf'],'contenttype','vector')
    exportgraphics(gcf,'sistemaME.png','Resolution',600);
    print('sistemaME','-dsvg');
    print('sistemaME','-depsc');
end
```